

Storage Devices

- Review of storage devices
- Why ROM is also considered as primary memory

Storage Devices:

- stores digital data, informations and instructions to be processed
- hardware that is used for storing, porting or extracting data files
- without storage devices computer not even start
- integral part of the computer hardware which stores information/data to process the result of any computational work.

Types of storage devices

- i) Primary storage Devices
- ii) Secondary storage Devices

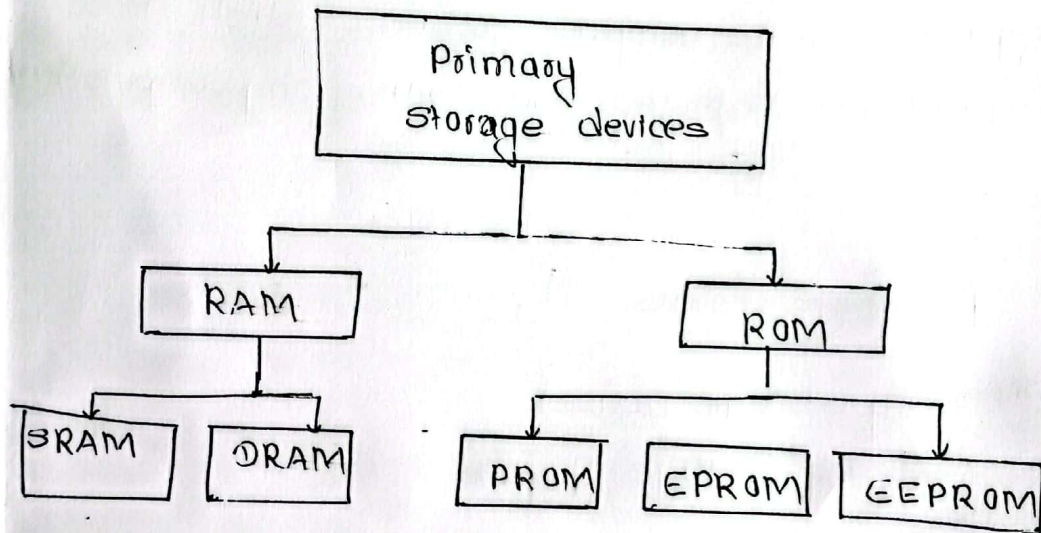
Assignment:

Primary memory Vs Secondary Memory

- * Primary storage Devices
- It is also known as main memory
- CPU access primary memory directly
- limited storage capacity
- fastest type of memory

- data lost when power is switched off
- It is generally made up of semiconductor devices
- Primary memory/storage devices is also known as main memory
- usually volatile in nature
- Primary storage devices is categorized in two two parts
 - RAM
 - ROM

* structure of Primary storage devices



RAM:

- RAM stands for Random Access Memory
- RAM is volatile in nature
- Alternatively RAM is also referred as main memory, primary memory, or system memory
- It allows information to be stored and retrieved on a computer
- It access the data much faster
- If the computer is off all data contained in RAM is lost

Types of RAM

i) SRAM

ii) DRAM

iii) SDRAM

iv) DDR SDRAM

v) RDRAM

SRAM:

- SRAM stands for Static Random Access Memory.
- It retains stored information only as long as the power supply is on.
- SRAM is more costlier in comparison to DRAM
- It consumed more power
- It has higher speed than DRAM
- No need to be refreshed
- RAM is used to create store data for short time
- It is composed of various electronic components like flip-flop

DRAM:

- DRAM (Dynamic Random Access Memory)
- It is cheaper than that of SRAM
- It consumed less power
- It is comparatively slower than SRAM
- Need to be refreshed
- It is composed of transistor and capacitor as an electronic component
- Data stored for very short time

Block diagram of RAM

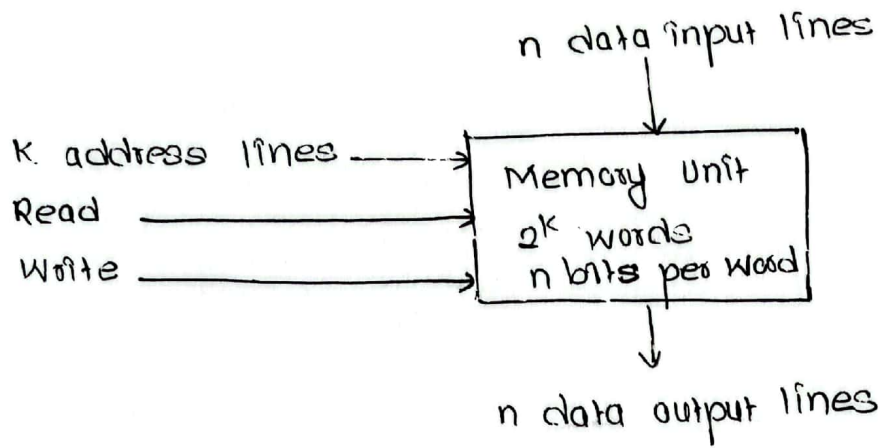


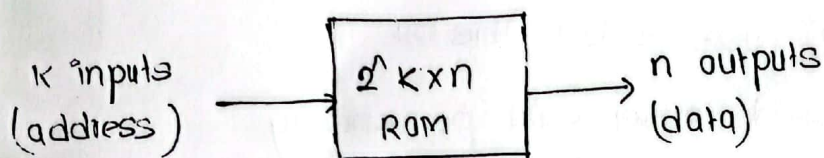
Fig: block diagram of RAM

The n data input lines provide the information to be stored in memory and n data output lines supply the information coming out of particular word chosen among the 2^k available inside the memory.

ROM:

ROM stands for Read Only Memory. It is a type of primary memory (main memory). ROM stores data permanently and cannot be modified that means ROM is immutable. ROM is non-volatile memory.

Any computer system must include ROM since it contains the instructions needed for computer to bootup and run its programs. ROM usually contains software or firmware that is rarely updated.



* Characteristics of ROM:

- It is primary memory
- It is non-volatile memory
- It store data permanently and data cannot modified
- ROM usually contains software or firmware
- Information stored in ROM is in binary format
- Without ROM computer not even start
-

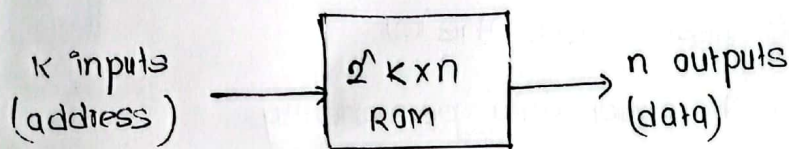
* Real field uses of ROM

→ Digital watches, microwaves, BIOS, smartphones, Embedded system

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Types of ROM:

- i) PROM
- ii) EPROM
- iii) EEPROM
- iv) Flash Memory/ROM
- v) Masked ROM

PROM:

- Programmable Read Only Memory
- The data is stored permanently.
- It cannot be erased and reprogrammed
- It uses fuses or anti fuses to lock the bit
- A high voltage is applied to set data permanently
- It is used in low level programs
eg microcode and firmware

EPROM:

- Erasable Programmable Read Only Memory
- It can be erased and reprogrammed ultraviolet light
- Reprogram is limited
- eg microcontroller
- It uses floating gate transistors

EEPROM:

- Electrically Erasable Programmable Read Only Memory
- It can be reprogrammed and erasable electrically
- Reprogrammed and erasable about ten thousand times
- To erase and program it takes 4ms to 10ms
- eg microcontroller, sensors

MROM:

→ Masked Read only memory

→ oldest ROM

→ During fabrication process, a specific pattern of transistor or diode created to store data permanently

→

Flash memory/Flash ROM:

→ built using floating gate transistor

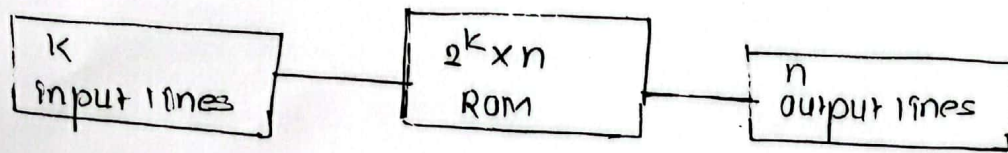
→ type of EEPROM

→ faster

→ large amount of data stored
smartphone

Secondary Memory

Block diagram of ROM



As shown in fig above there are K input lines and output lines in it. The input address from which we use to retrieve the ROM content take by K input lines. Each of the K input lines have value 0 or 1. There are total 2^K addresses that can be referred to by these input lines and each of these addresses contains n bits of information.

Internal structure of ROM

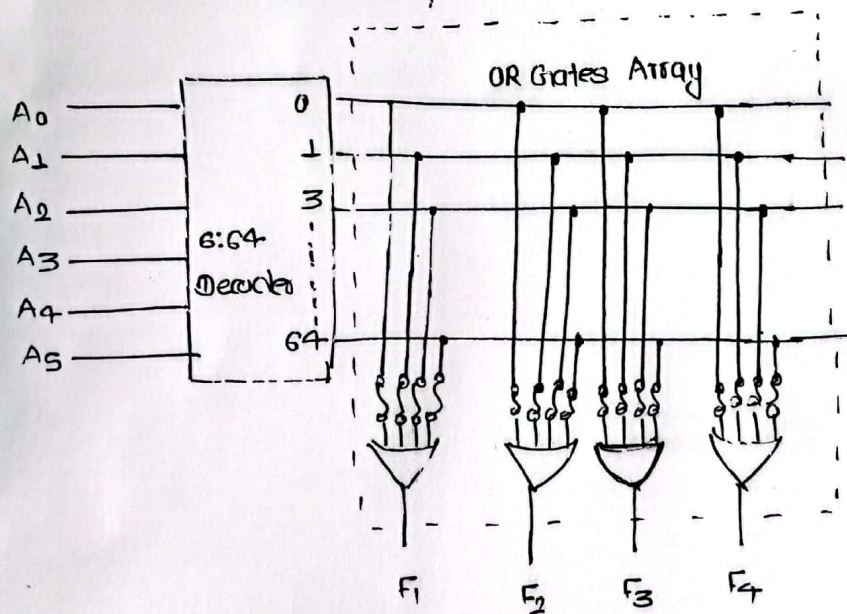


Fig: Internal Constriution of 64x4 ROM

A circuit known as decoder converts an encoded form, such as binary coded decimal or BCD into decimal form. So output is binary equivalent of the input. The output of the decoder will be output of every ~~or gate~~ OR gate in ROM.

Difference between RAM and ROM

RAM	ROM
RAM stands for Random Access Memory	i ROM stands for Read Only Memory
RAM is volatile memory it stores data as long as power supply is given	ii ROM is non-volatile memory that retain data even after the power is turned off.
It is faster than ROM	iii It is slower than RAM
It is comparatively costlier than ROM	iv It is cheap as compared to RAM
CPU can easily access data in RAM	v CPU cannot easily access data in ROM

Registers:

- Register is the smallest and fastest memory in computer
- Register is not considered as primary memory
- It is located inside the memory
- It holds frequently used data temporarily

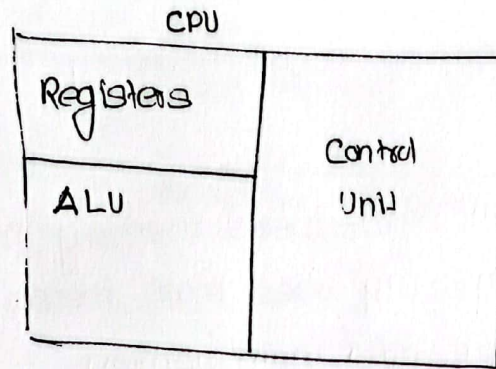


Fig: ~~CPU~~ Register inside CPU

Memory Hierarchy

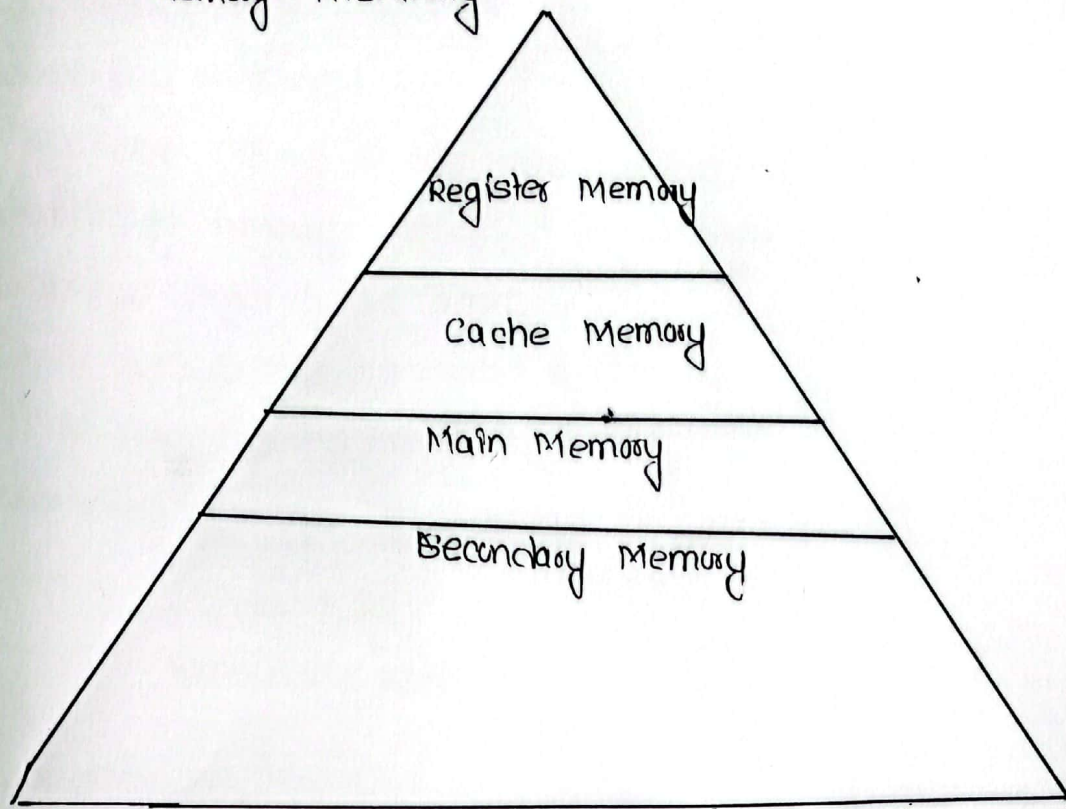


Fig: memory Hierarchy

Secondary Memory/Storage Devices:

Secondary Memory:

- Secondary Memory is also known as auxiliary memory
- Secondary memory stores data permanently
- storage capacity is high
- it is known as non-volatile memory
- It is not directly access by CPU
- Secondary storage devices can be fixed and removable

fixed - HDD

removable - pendrive

- fixed storage device like hard disk that are fixed inside the computer
- removable storage device like pendrive that can be removable and portable.
- data remain when the computer is switched off
- used for long term data storage, backup and transfer

What is Secondary Memory?

Memory in a computer refers to the physical components that are used to temporarily or permanently store programmes or data. It's a collection of registers.

Primary memory is very volatile and has a finite amount of storage space. As a result, having a second type of memory with a bigger storage capacity- from which programs and data are not lost whenever the computer is turned off is critical. Secondary memory is the term for this form of memory. Programs and data are stored in secondary memory. Auxiliary memory is another name for it. It differs from primary memory in that it is non-volatile and not directly accessible by the CPU. Secondary storage devices, often known as external storage devices, have substantially more storage space and are less expensive than primary memory.

Use of Secondary Memory

Secondary memory is mainly used for different purposes. The primary purpose of using it is as follows:

- 1. Permanent storage** – Primary memory, as we all know, retains data only when the power is on and loses data when the power is turned off. As a result, we require a secondary memory that can keep data indefinitely even if the power supply is turned off.
- 2. Large storage** – Secondary memory provides a significant amount of storage space, allowing us to store large amounts of data such as films, photographs, audio files, and other items permanently.
- 3. Portable** – Some of the secondary devices can be removed. As a result, one can store data and transfer them effortlessly from one device to another.

Types of Secondary Memory

There are two types of secondary memory:

Types of Removable Storage:

Here are the types of removable storage:

- Optical discs (such as DVDs, CDs, Blu-ray discs, etc.)
- Floppy disks
- Memory cards
- Disk packs
- Magnetic tapes
- Paper storage (such as punched cards, punched tapes, etc.)

Secondary Memory Devices

Here are the most commonly utilised secondary memory devices:

1. Floppy Disk – A floppy disc refers to a magnetic disc housed in a square plastic container. It is primarily used to store information and then transfer it from one device to another. There are two sizes of floppy discs (a) Size: 3.5 inches, 1.44 MB storage capacity (b) Size: 5.25 inches and 1.2 MB storage capacity. A computer must consist of a floppy disc drive for using a floppy disc. Such a storage device is now obsolete, and DVDs, CDs, and flash drives have taken its place.

2. Compact Disc – A Compact Disc or the CD is a type of secondary storage media that is widely utilised. Its surface is covered in tracks and sectors. It is composed of polycarbonate plastic and has a round form. CDs have a data storage capacity of up to 700 MB. A CD is also known as a CD-ROM (Compact Disc Read-Only Memory) because computers can read the data on it but not write new data onto it.

There are two types of CDs:

- **CD-R or compact disc recordable** – It is not possible to remove data that has been written onto it; it can only be read.
- **CD-RW or compact disc rewritable** – It's a particular kind of CD that allows us to wipe and rewrite data as many times as we desire, so it's also known as an erasable CD.

3. Digital Versatile Disc – DVD stands for Digital Versatile Disc. It resembles a CD, but it has a larger storage capacity; up to 4.7 GB of data can be stored. To use a DVD on a computer, a DVD-ROM drive is required. Video files, such as movies or video recordings, are commonly saved on DVDs, which may be played with a DVD player. There are three different kinds of DVDs available:

- **DVD-ROM or Digital Versatile Disc Read-Only Memory** – The manufacturer writes the data on the DVD-ROM, and the users can only read it; they cannot create new data on it. For example, a movie DVD has already been written by the maker, and we can only watch it; we cannot add new data to it.
- **DVD-R or Digital Versatile Disc Recordable** – You can write data on a DVD-R, but only once. It is not possible to remove data that has been written onto it; it can only be read.
- **DVD-RW or Digital Versatile Disc RW (Rewritable and Erasable)** – It's a particular kind of DVD that allows us to wipe and rewrite data as many times as we desire. It's also known as an erasable DVD.

4. Blu-ray Disc – It resembles a CD/VD, but it can actually hold up to 25 GigaBytes data or information. You'll need a Blu-ray reader if you wish to use a Blu-ray disc. The name Blu-ray comes from the technology used to read the disc: 'Blu' comes from a blue-violet laser, and 'ray' comes from an optical ray.

5. Hard Disk – Hard discs are storage devices designed to hold a lot of information. Hard discs, often known as hard disc drives, are available in a variety of storage capacities. (For example, 256 GB, 500 GB, 1 TB, and 2 TB). It is made up of platters, which are a collection of discs. The dishes are stacked one on top of the other. They have a magnetic coating on them. Each platter is made up of a number of invisible circles, each of which has the same central track. Internal and external hard discs are the two types of hard discs.

6. Flash Drive – A flash drive or pen drive is available in a variety of storage capacities, including 1, 2, 4, 8, 16, 32, 64 GB, and 1 TB. Data is transferred and stored using a flash drive. To use a flash drive, it must be plugged into a computer's USB port. A flash drive is quite popular nowadays because it is simple to use and small in size.

7. Solid-state Disk – A Solid-state Disk (SSD) is a data storage and retrieval device that is non-volatile. It's faster, makes no noise (since it doesn't have any moving parts like a hard drive), uses less electricity, and so on. If the price is right, it's a perfect replacement for regular hard drives in laptops and desktops, and it's also ideal for tablets, notebooks, and other devices that don't require a lot of storage.

8. SD Card – A Secure Digital Card (SD Card) is commonly used to store data in portable devices such as cell phones and cameras. It comes in a variety of capacities, including 1, 2, 4, 8, 16, 32, and 64 GB. You can access the data present in the SD card by removing it from the device and inserting it into a computer using a card reader. The data in the SD card is stored in memory chips (which are included in the SD card), and there are no moving elements like there are on a hard disc.